

SQLite Subsystem — #d.u.m.b.a.s.s.

Purpose & Scope

The SQLite tier provides zero-latency, on-device structured storage and local caching across the federated ecosystem. Designed specifically for **Node Alpha** (Motorola RAZR Ultra 2025 / Termux / Æsc daemon) and local daemon processes on **Node Beta** (Jetson Orin Nano).

Key Components

1. **schema.sql**: The canonical local schema definition implementing:
 - **kv_store**: Microsecond key-value caching with namespace isolation.
 - **active_task_state**: Persistent state tracking allowing interrupted agent tasks to resume seamlessly without losing conversation context.
 - **entity_lookup**: Canonical naming, aliases, and fast entity resolution without hitting heavy vector search.
 - **local_token_cache**: Exact-match prompt/response cache supporting RTK compression and reducing redundant inference calls.
 - **episodic_event_queue**: Local staging buffer that captures edge events before P2P synchronization to the master PostgreSQL database on Node Beta.

Performance Tuning

- **WAL Mode** (`PRAGMA journal_mode = WAL;`): Ensures concurrent reads do not block background logging daemons.
- **Normal Sync** (`PRAGMA synchronous = NORMAL;`): Maximizes throughput on Android UFS 4.0 flash storage while maintaining integrity across app restarts.